AI-Powered Visual Data Analytics and Business Intelligence Platform

Milestone 1 Report: Workplace PPE Detection Using Computer Vision

Project Focus
Computer-vision-based workplace safety monitoring

Milestone 1 Scope
PPE detection in images and videos, annotated outputs, and CSV violation reports

1. Project Idea

The project aims to analyze workplace images and videos to identify Personal Protective Equipment (PPE) and flag potential safety violations. The initial milestone focuses on construction-site safety scenarios and converts visual detections into annotated media and structured CSV reports.

The application uses a YOLO-based object-detection model to recognize PPE-related classes such as Hardhat, Safety Vest, Gloves, Goggles, and Mask, together with corresponding missing-PPE labels. The system is intended to support safety review and reporting; model predictions are indicators for human review rather than definitive proof of a violation.

2. Milestone 1 Objectives

- Set up a local Python environment and install the required computer-vision packages.
- Load and test the existing PPE detection model (best.pt).
- Process multiple images and save annotated output images.
- Process videos frame by frame and save annotated MP4 output.
- Record potential PPE violations, confidence scores, and video timestamps in CSV reports.
- Prepare the foundation for a web-based visual analytics interface.

3. Emerging Skills

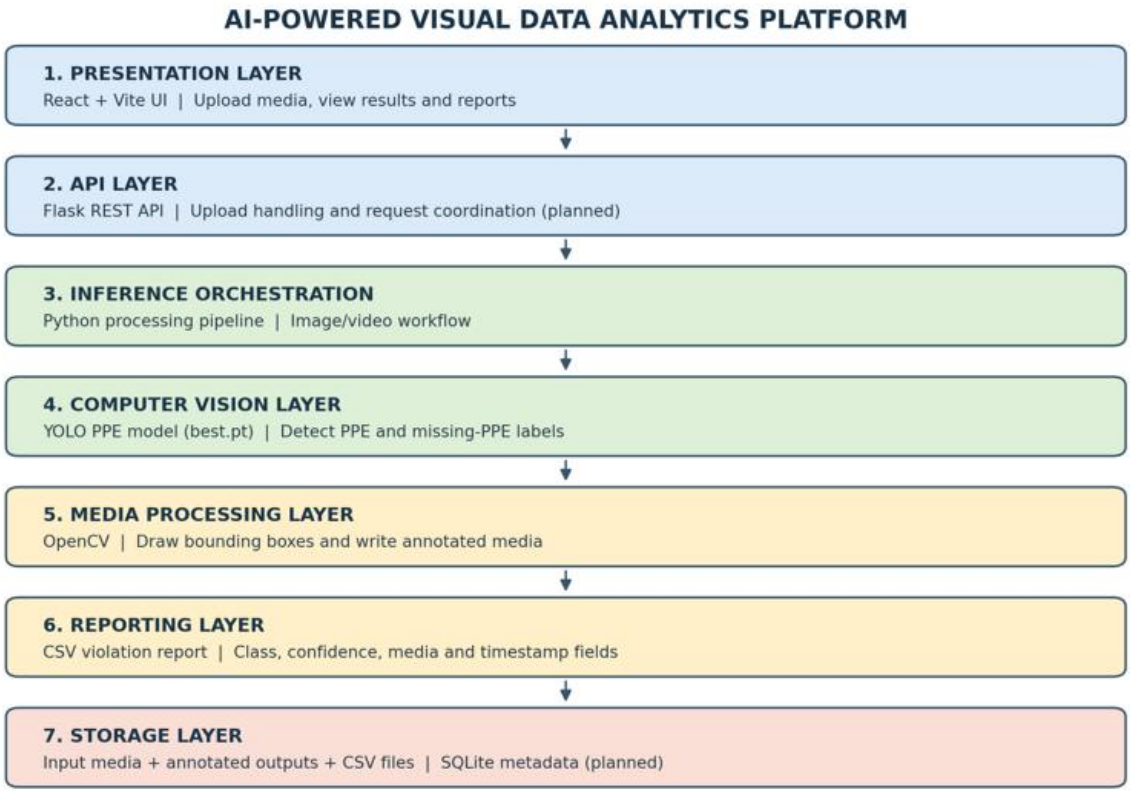
- Python scripting and virtual environments
- Computer vision fundamentals and object detection
- YOLO inference and confidence thresholds
- OpenCV image/video reading, annotation, and writing
- CSV report generation and file handling
- Debugging Python dependencies and Windows terminal workflows

4. Technology Stack

Component	Technology	Purpose
AI / Computer Vision	Ultralytics YOLOv8-compatible model	Detects PPE classes and potential missing-PPE labels
Model file	best.pt	Existing PPE detection weights
Image / Video Processing	OpenCV (cv2)	Reads media, draws detections, writes annotated outputs
Language	Python	Inference workflow and report generation
Reporting	CSV	Stores violation labels, confidence, and video timestamps
Frontend (application direction)	React + Vite	Interactive user interface
Backend (application direction)	Flask REST API	Upload handling and model integration
Charts (application direction)	Recharts	Dashboard and analytics visualizations
Database (application direction)	SQLite	User accounts and analysis history, subject to final implementation

5. System Architecture

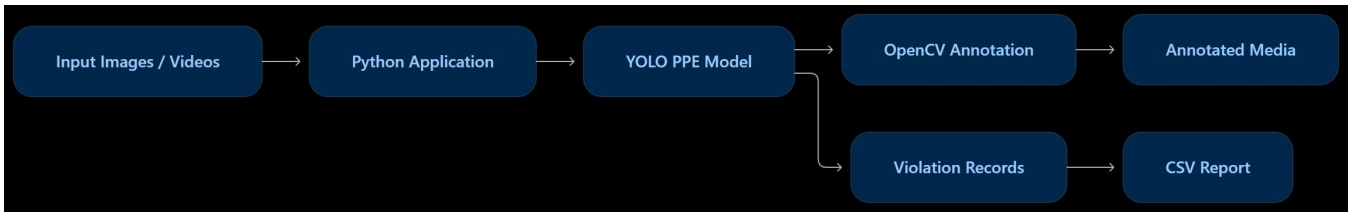
The implemented computer-vision workflow is a local Python pipeline. Images and videos are supplied from an input folder; the model predicts object classes; OpenCV produces annotated media; and CSV files record potential violations. The planned full-stack application places a React interface in front of a Flask API and can persist analysis metadata in SQLite.



6. Project Working Diagram

Confirmed local workflow: Input media → YOLO model (best.pt) → OpenCV annotation → CSV violation report.

Target full-stack workflow: React UI ↔ Flask REST API ↔ YOLO/OpenCV; analysis metadata and detections can be stored in SQLite; annotated media and CSV reports are returned to the user.



7. Class / Data Model

Entity	Key fields	Relationship / purpose
User (if implemented)	id, name, email, password_hash	Owns analysis records
Analysis	id, user_id, filename, media_type, status, timestamps, output_path, report_path	One uploaded media item and its processing result
Detection	id, analysis_id, class_label, confidence, frame_number, time_seconds	Individual model detection; many detections can belong to one analysis

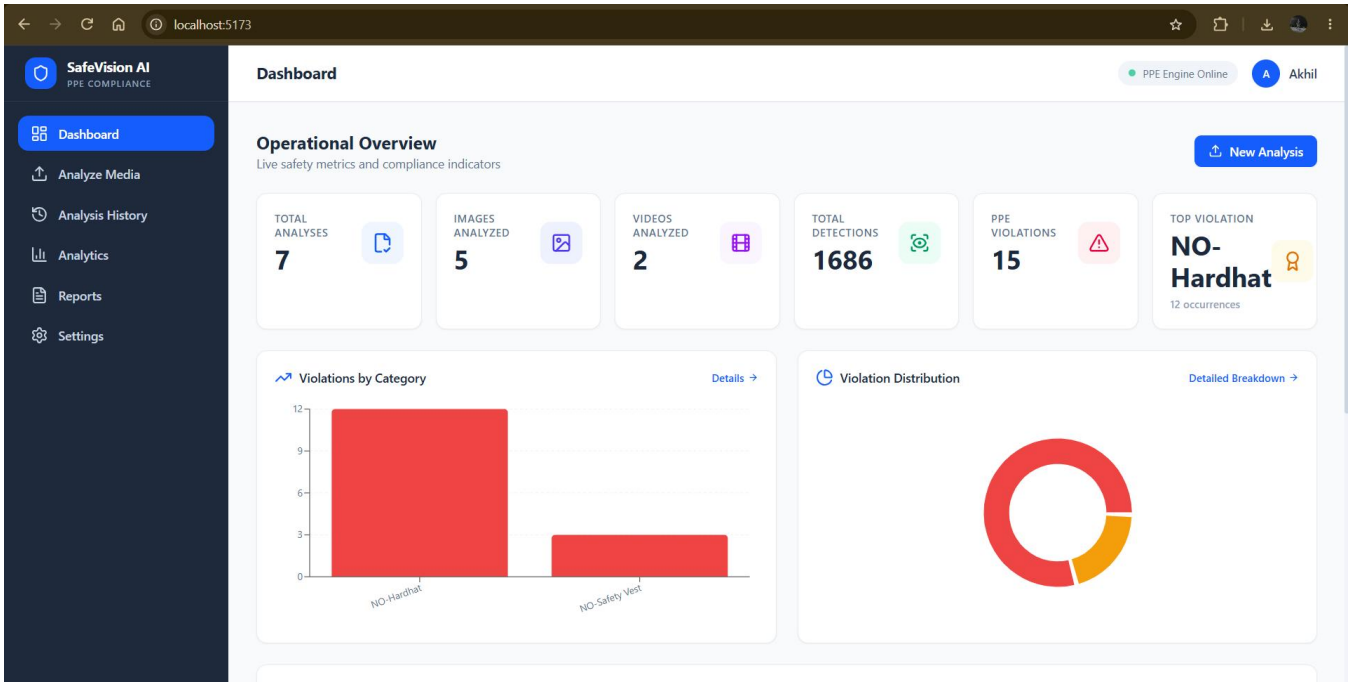
Logical data model only; verify names and fields against the final database implementation.

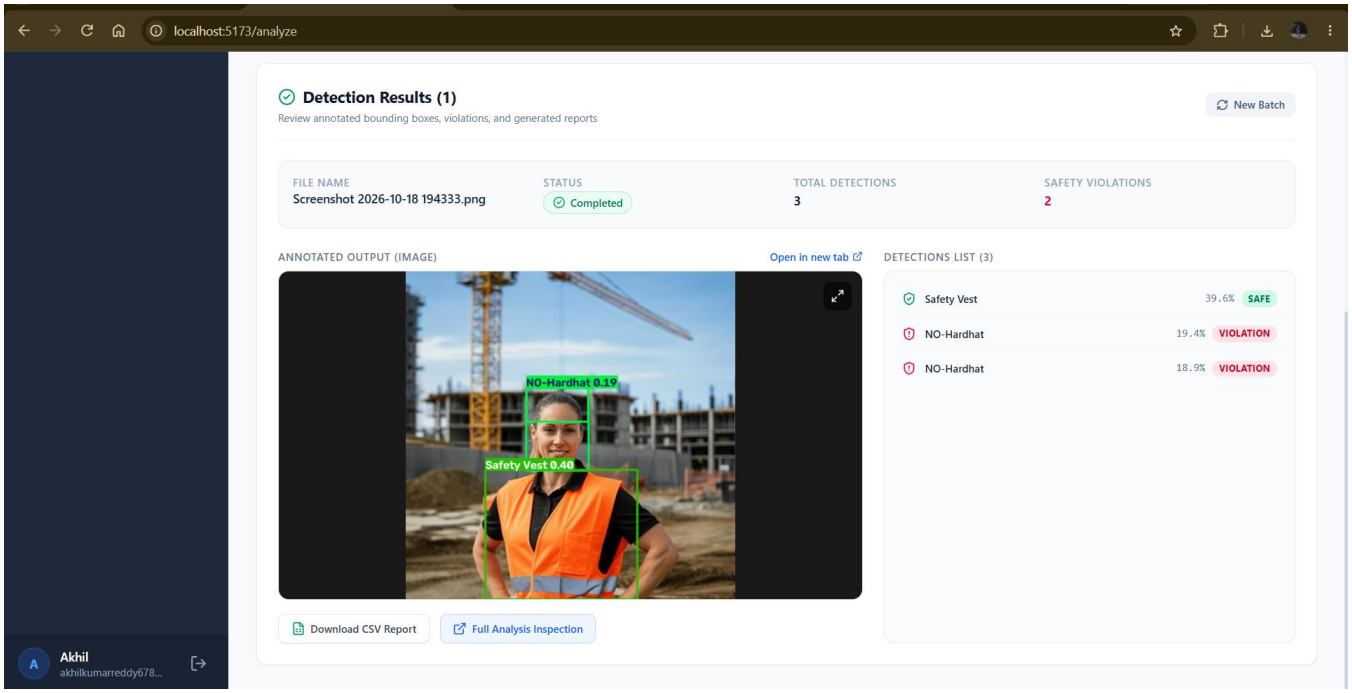
8. Milestone 1 Deliverables (Weeks 1-2)

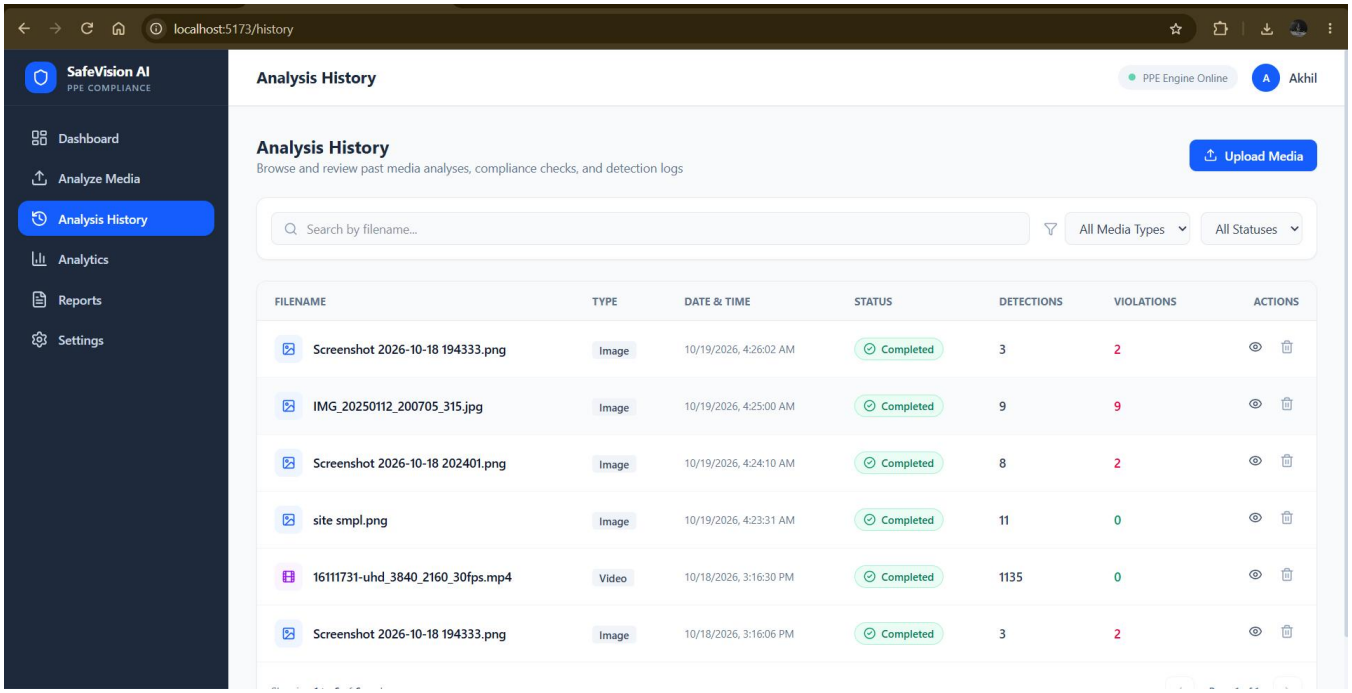
- PPE model loaded locally from best.pt.
- Image inference through detect.py.
- Annotated image outputs saved under outputs/.
- Video inference through video_detect.py.
- Annotated MP4 output generated from video frames.
- CSV report generation for potential PPE violations.
- Multiple-image processing and multiple-video processing support in the scripts.
- Local environment and dependency troubleshooting completed.

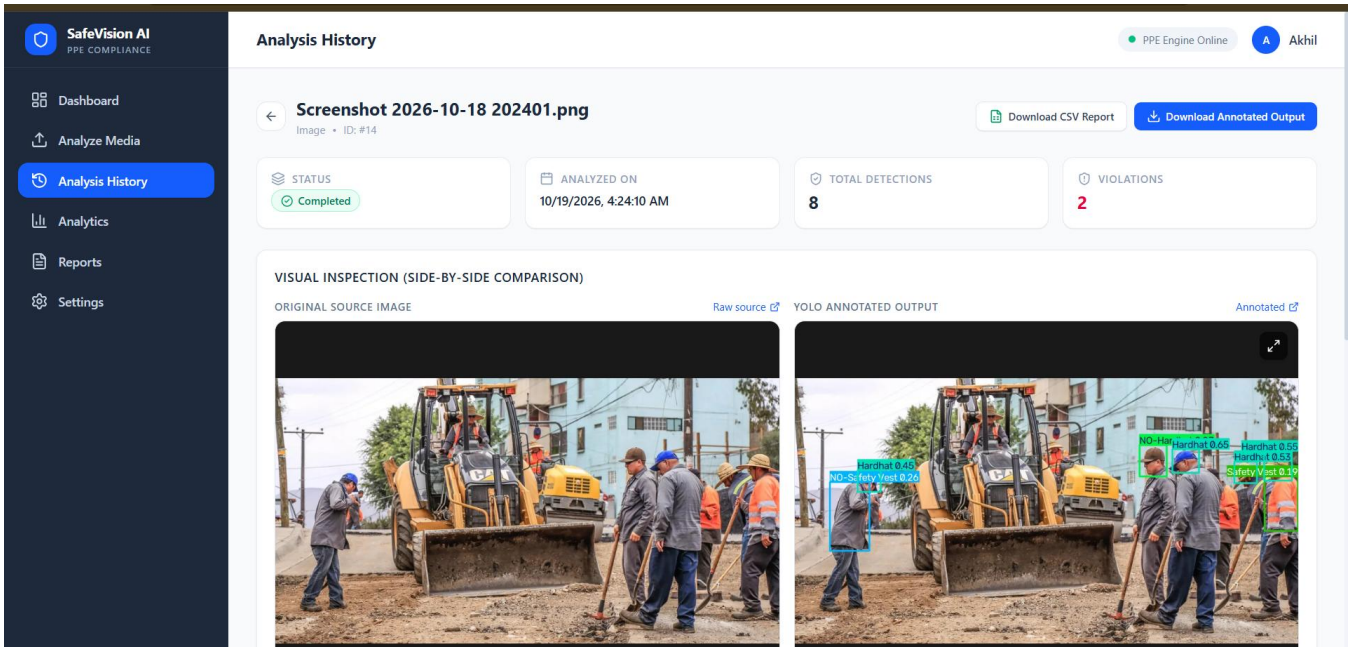
9. Expected Output Screens

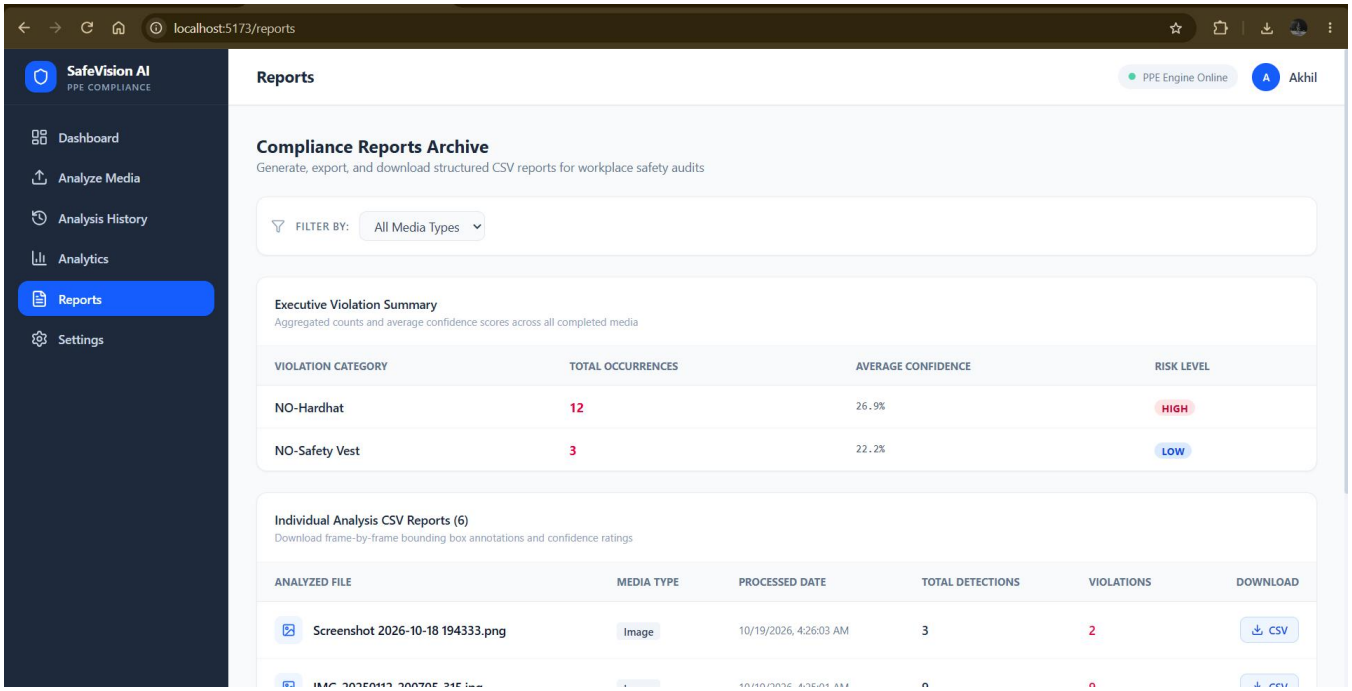
The intended full-stack interface follows a modern analytics dashboard design. Screens include Login/Register, Dashboard, Analyze Media, Image Results, Video Results, Analysis History, Analytics, and Reports. These are the target application screens; actual screenshots should be inserted after capturing the final running application.

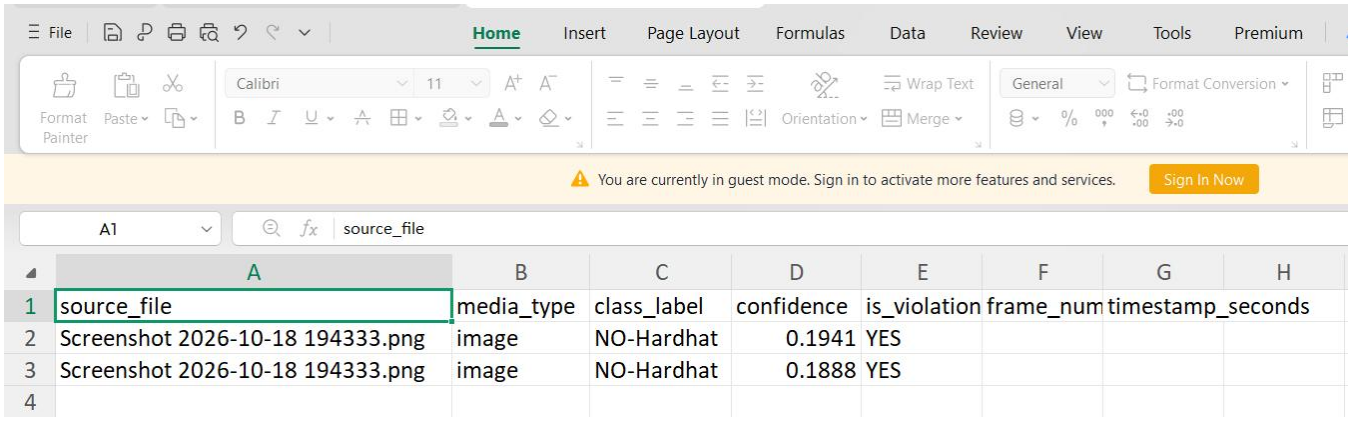

localhost:5173
SafeVision AI
PPE COMPLIANCE
Dashboard
Analyze Media
Analysis History
Analytics
Reports
Settings
Dashboard
PPE Engine Online
A
Akhil
Operational Overview
Live safety metrics and compliance indicators
New Analysis
TOTAL ANALYSES
7
IMAGES ANALYZED
5
VIDEOS ANALYZED
2
TOTAL DETECTIONS
1686
PPE VIOLATIONS
15
TOP VIOLATION
NO-Hardhat
12 occurrences
Violations by Category
Details →
12
9
6
3
0
NO-Hardhat
NO-Safety Vest
Violation Distribution
Detailed Breakdown →


localhost:5173/analyze
Detection Results (1)
Review annotated bounding boxes, violations, and generated reports
New Batch
FILE NAME
Screenshot 2026-10-18 194333.png
STATUS
Completed
TOTAL DETECTIONS
3
SAFETY VIOLATIONS
2
ANNOTATED OUTPUT (IMAGE)
Open in new tab
NO-Hardhat 0.19
Safety Vest 0.40
DETECTIONS LIST (3)
Safety Vest 39.6% SAFE
NO-Hardhat 19.4% VIOLATION
NO-Hardhat 18.9% VIOLATION
Download CSV Report
Full Analysis Inspection
Akhil
akhilkumarreddy678...


localhost:5173/history
SafeVision AI
PPE COMPLIANCE
Dashboard
Analyze Media
Analysis History
Analytics
Reports
Settings
Analysis History
PPE Engine Online
A
Akhil
Analysis History
Browse and review past media analyses, compliance checks, and detection logs
Upload Media
Search by filename...
All Media Types
All Statuses
FILENAME
TYPE
DATE & TIME
STATUS
DETECTIONS
VIOLATIONS
ACTIONS
Screenshot 2026-10-18 194333.png
Image
10/19/2026, 4:26:02 AM
Completed
3
2
IMG_20250112_200705_315.jpg
Image
10/19/2026, 4:25:00 AM
Completed
9
9
Screenshot 2026-10-18 202401.png
Image
10/19/2026, 4:24:10 AM
Completed
8
2
site smpl.png
Image
10/19/2026, 4:23:31 AM
Completed
11
0
16111731-uhd_3840_2160_30fps.mp4
Video
10/18/2026, 3:16:30 PM
Completed
1135
0
Screenshot 2026-10-18 194333.png
Image
10/18/2026, 3:16:06 PM
Completed
3
2


SafeVision AI
PPE COMPLIANCE
Dashboard
Analyze Media
Analysis History
Analytics
Reports
Settings
Analysis History
PPE Engine Online
A
Akhil
Screenshot 2026-10-18 202401.png
Image • ID: #14
Download CSV Report
Download Annotated Output
STATUS
Completed
ANALYZED ON
10/19/2026, 4:24:10 AM
TOTAL DETECTIONS
8
VIOLATIONS
2
VISUAL INSPECTION (SIDE-BY-SIDE COMPARISON)
ORIGINAL SOURCE IMAGE
Raw source
YOLO ANNOTATED OUTPUT
Annotated
Hardhat 0.45
NO-Safety Vest 0.26
Hardhat 0.65
Hardhat 0.55
Hardhat 0.53
Safety Vest 0.19





10. Validation and Observations

A video-processing run completed successfully after the virtual environment was activated. The run processed more than 240 frames and recorded 13 potential violation detections. This is a count of frame-level detections, not 13 unique incidents.

A previous image test detected labels including NO-Safety Vest and NO-Hardhat and wrote them to a CSV report. Confidence values are model confidence scores; they do not represent overall model accuracy.

12. Conclusion

Milestone 1 established a working foundation for PPE detection in images and videos, including annotated outputs and CSV-based violation reporting. This foundation can be integrated into a full-stack visual analytics application, with dashboard metrics and reports derived from actual detection records.